

Key takeaways

Cisco devices have two types of configurations:

- The running configuration: the one that is currently active (stored in NVRAM).
- The startup configuration: the one the device loads while booting up (stored in flash memory).

Basic configuration commands:

- Set the hostname using the **hostname {name}** command.
- Set a password to the privileged mode using **enable secret** command.
- Reload a Cisco device using the **reload** command.
- Display the running configuration using the **show running-config** command.

IP Address and subnet mask:

- An IP address is formatted in dotted decimal format. Examples:
 - 192.168.1.1
 - 10.1.1.1
- A subnet mask determines the network or subnet to which a device belongs to.
Examples:
 - 255.255.255.0
 - 255.255.0.0
- IP addresses can be assigned to VLANs on Cisco switches.
- To view IP address assignment on a Cisco device, use the **show ip interface brief** command.
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Testing end-to-end connectivity:

- Test whether another device is reachable using the **ping {ip-address}** command.

Configuring DHCP servers:

- On the gateway router, define a DHCP pool using the **ip dhcp pool {pool-name}** command.
- Define the IP addresses range using the **network {net-id} {subnet-mask}** command.
- Specify a default gateway address using the **default-router {gateway-address}** command.
- Exclude the IP address of the router itself from the DHCP assignment by running the **ip dhcp excluded-address** command from the global configuration mode.